IMPERIAL

GAME THEORY AND MECHANISM DESIGN

IMPERIAL COLLEGE LONDON

DYSON SCHOOL OF DESIGN ENGINEERING

Cournot and Stackelberg Games

Author:

[redacted]

20 March, 2025

Task Allocation Table

Task	**Nora Luo**	**Site Ma**
Game description	50%	50%
Cournot game analysis	60%	40%
Stackelberg game analysis	40%	60%
Solution/Outcome discussion	50%	50%
Coding, visualisation and discussion	30%	70%
Extensions	50%	50%
Games in the real-world & validation	70%	30%
Quality of the report	50%	50%

I. Introduction

Oligopolistic markets are characterized by strategic interactions among firms, where decisions by one firm significantly influence by the actions of competitors. This report analyzes such interactions through two fundamental models: Cournot (simultaneous decisions) and Stackelberg (sequential decisions), emphasizing their strategic and welfare implications. The report explores how differences in firms' cost structures influence market efficiency, and also extend the framework to markets with multiple competitors, and investigate the circumstances under which firms might choose collusion over competition. A case study about the competition between airlines and their alliances was explored to examine the application of the theory.

II. Background

Oligopoly theory originated with the influential work of Antoine Augustin Cournot in 1838, which modeled mathematically of the strategic interactions among firms. [1] Cournot introduced the concept of firms making interdependent output decisions, inspired by observing competition in a spring water duopoly [2]. Nearly a century later, Heinrich von Stackelberg expanded on Cournot by introducing sequential decision-making in 1934, highlighting how the timing of decisions influences market outcomes and giving rise to the concept of first-mover advantage [3].

A. Cournot Competition

The Cournot game describes an oligopolistic market where firms simultaneously choose their production quantities. The market price is determined endogenously through market clearing by the balance of supply and demand. This aligns with Nash's non-cooperative equilibrium, where each firm selects its best response to the fixed output of rivals [4]. A key assumption is that firms treat rivals' output as fixed ("conjectural variation" of zero). Along side discussions on its consistency and applicability [5], this assumption remains widely used due to its analytical simplicity and applicability in markets characterized by rigid capacities and significant initial investments.

B. Stackelberg Competition

The Stackelberg model introduces a sequential dimension, with one firm acts first as the "leader", committing to a production quantity, and the remaining firms respond afterward as "followers". This sequential structure results in a subgame perfect equilibrium, requiring the strategies to be optimal at each stage of decision-making. The leader may anticipate followers' reactions and influence the market dynamics. Such credible commitment typically depends on observable and irreversible actions, including capacity investments or technological leadership [6]. Thus, the Stackelberg model differs from Cournot by highlighting strategic credibility beyond timing.

C. Market Structure and Assumptions

The report analyses a duopolistic market in a dystopian future where water is traded as a scarce resource. The market has the following characteristics:

1) **Homogeneous Product:** Firms produce identical, substitutable products (water), leading to a single market price.
2) **Linear Demand:** Consumer demand is elastic and linearly related to price, defined as:

$$Q = \alpha_d - \beta_d p, \quad \alpha_d, \beta_d > 0 \tag{1}$$

, where Q is total quantity demanded, p is the price, α_d represents the maximum potential demand, and β_d is the price sensitivity.
3) **Constant Marginal Costs:** Firms A and B face linear cost structures with constant marginal costs:

$$c(q_1) = \beta_1 q_1, \quad \beta_1 > 0 \tag{2}$$

$$c(q_2) = \beta_2 q_2, \quad \beta_2 > 0 \tag{3}$$

, where q_1 and q_2 denote the quantity produced by firm A and B respectively. β_1 and β_2 represent the constant marginal costs of producing water, highlighting the cost differences and efficiency asymmetry.
4) **Complete Information:** All parameters and strategies are common knowledge among firms.
5) **Profit Maximization:** Firms rationally aim to maximize their own profits.

From equation (1), we can derive the price (p) as a function of quantities (Q), also known as the inverse demand curve.

$$p = \frac{\alpha_d - Q}{\beta_d} \tag{4}$$

The total quantity supplied to the market by both firms is:

$$Q = q_1 + q_2 \tag{5}$$

The numerical analysis in this report uses the following parameters: $\alpha_d = 100$, $\beta_d = 0.5$, $\beta_1 = 0.5$, $\beta_2 = 0.1$

III. Equilibria Analysis

A. Profit Functions and Strategic Interaction

Each firm's profit (π) can be described as the revenue generated by selling q_1 of products at price p deducted by the cost as expressed in equation (2)

$$\pi_1 = pq_1 - \beta_1 q_1 \tag{6}$$

$$\pi_2 = pq_2 - \beta_2 q_2 \tag{7}$$

By substituting inverse demand function equation (4) to equation (6)(7), the profit become functions of variables q_1 and q_2:

$$\pi_1 = \frac{\alpha_d q_1 - q_1^2 - q_1 q_2}{\beta_d} - \beta_1 q_1 \tag{8}$$

$$\pi_2 = \frac{\alpha_d q_2 - q_2^2 - q_1 q_2}{\beta_d} - \beta_2 q_2 \tag{9}$$

These equations reflect the strategic interdependence between firms through the interaction term $q_1 q_2$. The negative cross-partial derivatives:

$$\frac{\partial^2 \pi_i}{\partial q_i \partial q_j} = -\frac{1}{\beta_d} < 0$$

confirm that the quantities are strategic substitutes. Thus, each firm's output increase decreases the rival's marginal profit, incentivizing output reduction.

B. Cournot Equilibrium

In Cournot competition, firms simultaneously select outputs. Since the second derivative $\frac{\partial^2 \Pi_i}{\partial q_i^2} = -\frac{2}{\beta_d} < 0$ indicates that the profit function is strictly concave, setting the marginal profit (first derivative) equal to zero identifies a maximum, thus representing the optimal decision (best response) of each firm given the rival's output. Firm A's marginal profit set to zero:

$$\frac{\partial \pi_1}{\partial q_1} = \frac{\alpha_d - 2q_1 - q_2}{\beta_d} - \beta_1 = 0 \tag{10}$$

Noticing the first term in equation above is the marginal revenue (MR_1) for Firm A and the second term is its marginal cost (MC_1). Therefore, Firm A's best response function is:

$$q_1^*(q_2) = \frac{\alpha_d - q_2 - \beta_1 \beta_d}{2} \tag{11}$$

Since the condition is symmetrical for Firm B, the best response function for Firm B is:

$$q_2^*(q_1) = \frac{\alpha_d - q_1 - \beta_2 \beta_d}{2} \tag{12}$$

Te Nash Equilibrium is reached when are playing mutual best response to one another. There is no incentive for either firm to change from this equilibrium. The Cournot-Nash equilibrium is the intersect of these negatively sloped reaction functions:

$$q_1^C = \frac{\alpha_d + \beta_2 \beta_d - 2\beta_1 \beta_d}{3} \tag{13}$$

$$q_2^C = \frac{\alpha_d + \beta_1 \beta_d - 2\beta_2 \beta_d}{3} \tag{14}$$

The total market output is:

$$Q^C = \frac{2\alpha_d - (\beta_1 + \beta_2)\beta_d}{3} \tag{15}$$

and the equilibrium price is

$$p^C = \frac{\alpha_d + (\beta_1 + \beta_2)\beta_d}{3\beta_d} \tag{16}$$

This equilibrium analysis offers several key insights into duopolistic competition. It shows that each firm's equilibrium output is inversely related to its own marginal cost and positively affected by its rival's marginal cost, underscoring the cost asymmetry effects. Moreover, a larger market size, reflected in a higher α_d, increases total production for both firms, while greater demand sensitivity (β_d) amplifies the impact of cost differences on production decisions.

C. Stackelberg Equilibrium

In the Stackelberg model, firms set outputs sequentially. Firm A (the leader) chooses its output first, and Firm B (the follower) responds optimally. The solution is obtained by using backward induction, a method commonly applied in sequential games.

Since Firm B moves as the follower, and B knows the leader's quantity q_1, the best response function of the follower is identical to the Cournot game, same as equation (12):

$$q_2^*(q_1) = \frac{\alpha_d - q_1 - \beta_2 \beta_d}{2}$$

This function describes how Firm B optimally adjusts its quantity in response to any choice made by Firm A.

Firm A, as the leader, anticipates Firm B's reaction function (12). Its profit function (6) becomes,

$$\pi_1 = \frac{\alpha_d - (q_1 + q_2^*(q_1))}{\beta_d} q_1 - \beta_1 q_1$$

Substituting $q_2^*(q_1)$:

$$\pi_1 = \frac{\alpha_d - q_1 - \frac{\alpha_d - q_1 - \beta_2 \beta_d}{2}}{\beta_d} q_1 - \beta_1 q_1$$

Simplifying:

$$\pi_1 = \left(\frac{\alpha_d - q_1 + \beta_2 \, \beta_d}{2\beta_d} \right) q_1 - \beta_1 \, q_1 \tag{17}$$

Taking the first-order derivative with respect to q_1 and setting it to zero for Firm A's profit maximization, similarly to Cournot:

$$\frac{\partial \pi_1}{\partial q_1} = \frac{\alpha_d - 2q_1 - \beta_2 \beta_d}{2\beta_d} - \beta_1 = 0$$

Solving for q_1, we obtain:

$$q_1^S = \frac{\alpha_d + \beta_2 \, \beta_d - 2\beta_1 \, \beta_d}{2} \tag{18}$$

Substituting this into Firm B's reaction function (12) provides the follower's optimal quantity:

$$q_2^S = \frac{\alpha_d - 3\beta_2 \, \beta_d + 2\beta_1 \, \beta_d}{4} \tag{19}$$

Total market output and price under Stackelberg equilibrium are:

$$Q^S = q_1^S + q_2^S = \frac{3\alpha_d - \beta_2 \, \beta_d - 2\beta_1 \, \beta_d}{4} \tag{20}$$

$$p^S = \frac{\alpha_d + \beta_2 \, \beta_d + 2\beta_1 \, \beta_d}{4\beta_d} \tag{21}$$

This Stackelberg model demonstrates the first-mover advantage: the leader sets its output anticipating the follower's reaction. The equilibrium shows that the leader's decision depends on its own cost, the follower's cost, and market demand, which in turn shapes total output and the market price.

While the Stackelberg model assumes the leader credibly commits to a production quantity, in practice, the leader has an incentive to strategically overstate this commitment. By announcing a higher quantity, the leader could manipulate the follower into reducing its production further, anticipating

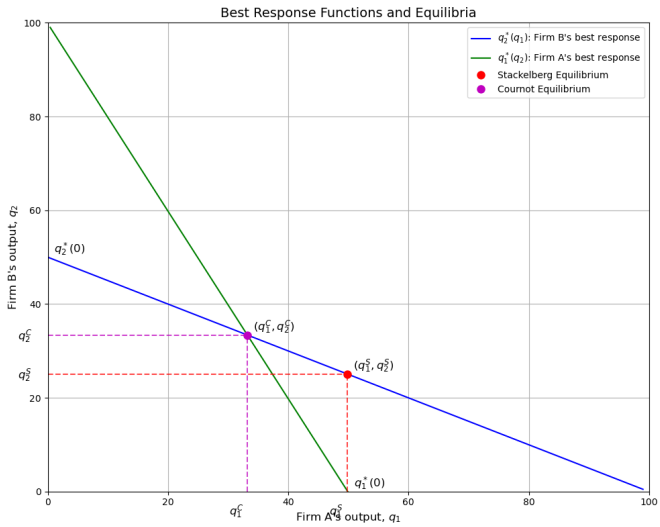


Fig. 1: Best responses and equilibria with given parameters

intensified market competition. However, such manipulation hinges critically on the credibility of the leader's announced quantity. If the follower suspects misrepresentation and disregards the leader's announcement, this strategic advantage collapses.

IV. Numerical Analysis

A. Equilibrium Solutions

Using the parameters in section II-C, we computed the equilibrium values for both Cournot and Stackelberg models.

1) Cournot Equilibrium: The best response functions are:

$$q_1^* = 50 - \frac{1}{2}q_2 - \frac{1}{8}$$

$$q_2^* = 50 - \frac{1}{2}q_1 - \frac{1}{40}$$

The calculated Cournot Equilibrium solutions under the given condition are:

Variable	Description	Value
q_1^C	Firm A's output	33.18
q_2^C	Firm B's output	33.38
Q^C	Total market output	66.57
p^C	Market price	66.87
π_1^C	Firm A's profit	2,202.27
π_2^C	Firm B's profit	2,228.89
Π^C	Joint industry profit	4,431.16

TABLE I: Market outcomes in Cournot competition

From Figure 1, it is observed that despite Firm A having higher marginal costs ($\beta_1 = 0.5$) compared to Firm B ($\beta_2 = 0.1$), the equilibrium quantities are relatively similar. This is because in Cournot competition, the strategic substitution effect (firms reduce output in response to rivals' increases) tends to balance the output differences that would otherwise be caused by cost asymmetry.

Variable	Description	Value
q_1^S	Leader's output (Firm A)	49.77
q_2^S	Follower's output (Firm B)	25.09
Q^S	Total market output	74.86
p^S	Market price	50.28
π_1^S	Leader's profit	2,477.55
π_2^S	Follower's profit	1,258.77
Π^S	Joint industry profit	3,736.32

TABLE II: Market outcomes in Stackelberg competition

2) Stackelberg Equilibrium: In the Stackelberg model, with firm A as the leader, the market outcomes become:

The Stackelberg equilibrium reveals an asymmetry compared to the Cournot outcome. Despite its higher marginal cost, Firm A leverages its leading position to secure nearly double the output and profit of Firm B. Sequential decision making alters the competitive dynamics beyond what can be determined by the cost structure.

B. Comparison of Cournot and Stackelberg Equilibria

1) Market Output Comparison: The Stackelberg model illustrates how a firm can strategically leverage its first-mover advantage to secure a more favorable market outcome. The first-mover advantage can be quantified by the formula:

$$\pi_1^S - \pi_1^C = \frac{((\alpha_d - \beta_1\beta_d) - \beta_2\beta_d)^2}{16}$$

The equilibria were compared in the following table (III). This output shift illustrates the strategic commitment value of the first-mover position. By committing to a higher output level, the leader effectively claims a larger market share while reducing the profitability of the follower's potential production.

Variable	Cournot	Stackelberg	Difference
Firm A's Quantity q_1	33.18	49.83	$+50.2\%$
Firm B's Quantity q_2	33.38	25.06	-24.9%
Total Quantity Q	66.57	74.89	$+12.5\%$
Price p	66.87	50.23	-24.9%
Firm A's Profit P_1	2202.27	2477.55	$+12.5\%$
Firm B's Profit P_2	2228.89	1256.26	-43.6%

TABLE III: Comparison of Cournot and Stackelberg Equilibria

2) Social Welfare Comparison: The social welfare is measured by the sum of consumer surplus and producer surplus [7]. Consumer surplus is the difference between the price and their maximum willingness to pay. It is calculated by,

$$CS = \frac{1}{2}(p_{\max} - p^*)Q^*$$

$$p_{\max} = \frac{\alpha_d}{\beta_d}$$

indicating the welfare for the consumers. The producer surplus is the sum of firms' profit, which is,

$$PS = \sum_{i=1}^{n} \Pi_i$$

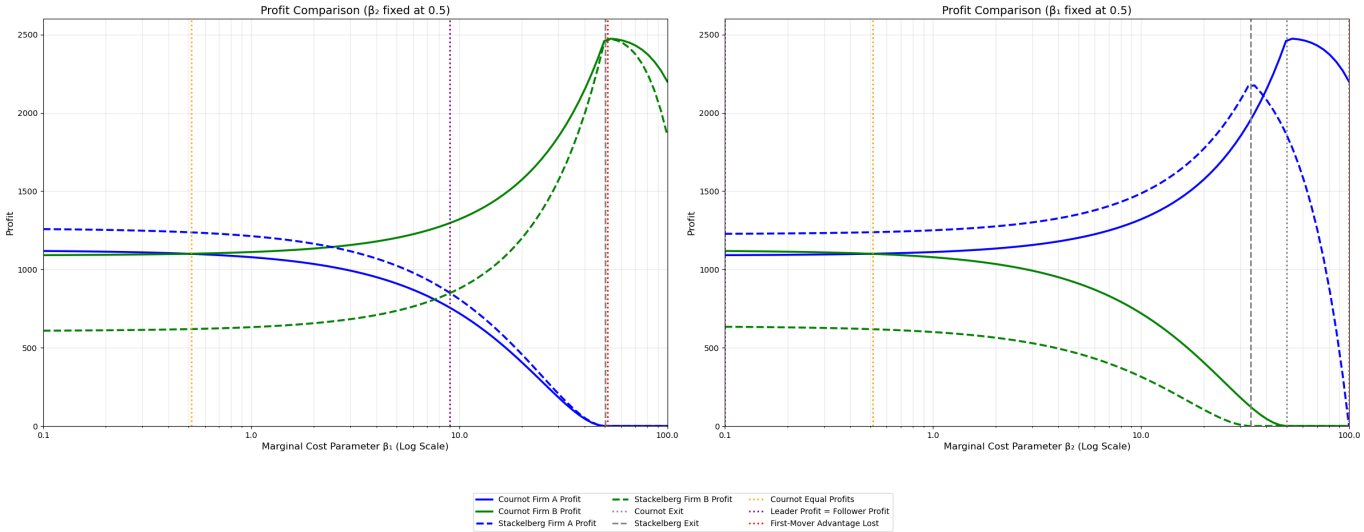


Fig. 2: Profit Comparison in Cournot and Stackelberg Competition with Varying Marginal Costs

The total welfare is the sum of CS and PS, reflecting the social welfare created by the market.

Welfare Measure	Cournot	Stackelberg	Difference
Consumer Surplus	1,109.18	1,401.59	+26.36%
Producer Surplus	4,431.16	3,736.32	−15.68%
Total Welfare	5,540.34	5,137.91	−7.26%

TABLE IV: Welfare Measures for Cournot and Stackelberg Models

As shown in Table IV, while Stackelberg competition increases competitive intensity and benefits consumers through lower prices and greater availability, the total welfare is reduced due to production inefficiency caused by the asymmetric output distribution between firms. Additionally, from a production efficiency standpoint, Stackelberg competition reverses the optimal cost-based allocation where the low-cost firm should produce more.

C. Cost Structure Sensitivity

Figure 2 compares profits under Cournot (solid lines) and Stackelberg (dashed lines) as firms' marginal costs vary. In the left panel, Firm A enjoys a significant first-mover advantage when its marginal cost (β_1) is low, maintaining higher Stackelberg profits compared to Cournot until $\beta_1 = 8.40$. Beyond this point, Firm A's advantage diminishes. Importantly, Firm A reaches the zero-profit exit threshold ($\beta_1 = 50.25$) before entirely losing its first-mover advantage. The right panel shows that Firm B consistently earns lower profits as a Stackelberg follower than under Cournot, underscoring the inherent strategic disadvantage for followers in sequential competition, especially when marginal costs are low.

V. Monopoly to Perfect Competition: Number of Agents in Cournot Game

While the basic Cournot model provides insights into duopolistic competition, real markets often feature more than two firms with varying cost structures. This extension explores how increasing the number of participants (n) affects equilibrium outcomes when firms have heterogeneous marginal costs.

A. Theoretical Extension

In the n-participant game with heterogeneous costs, each firm i chooses output q_i to maximize profit:

$$\Pi_i = pq_i - \beta_i q_i = \left(\frac{\alpha_d - Q}{\beta_d}\right) q_i - \beta_i q_i \tag{22}$$

where $Q = \sum_{j=1}^{n} q_j$ is the total market output.

The first-order condition for firm i yields its best response function:

$$q_i = \frac{\alpha_d - \beta_i\beta_d - \sum_{j\neq i}^{n} q_j}{2} \tag{23}$$

Solving this system of equations gives the Nash equilibrium quantity:

$$q_i^* = \frac{\alpha_d - n\beta_i\beta_d + \beta_d \sum_{j\neq i}^{n} \beta_j}{n+1} \tag{24}$$

Since production cannot be negative, firms with $q_i^* < 0$ will exit the market, meaning they become inactive and set $q_i^* = 0$. Therefore, the effective number of active firms n^* is dynamically determined by excluding those with negative equilibrium quantities.

The total market output and equilibrium price, accounting for active firms only, are:

$$Q^* = \sum_{i\in\mathcal{A}} q_i^* = \frac{n^*\alpha_d - \beta_d \sum_{i\in\mathcal{A}} \beta_i}{n^*+1} \tag{25}$$

$$p^* = \frac{\alpha_d + \beta_d \sum_{i\in\mathcal{A}} \beta_i}{(n^*+1)\beta_d} \tag{26}$$

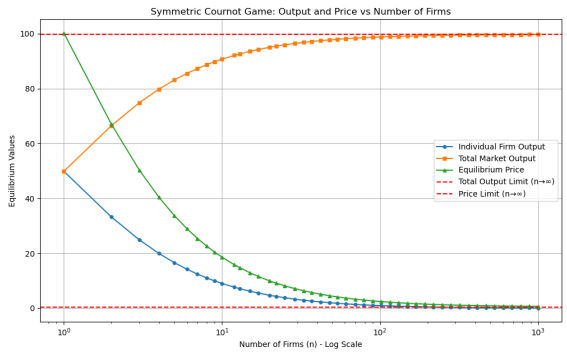


Fig. 3: Cournot market outcomes with homogeneous cost

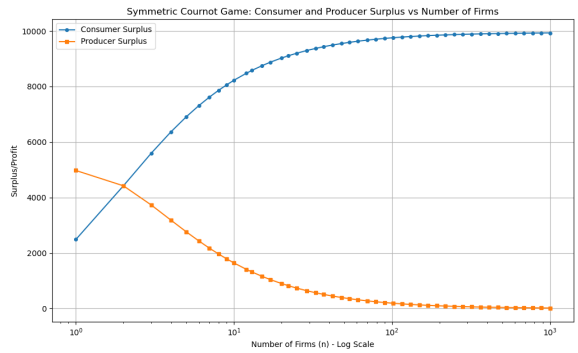


Fig. 4: Social welfare in Cournot game with homogeneous cost

where $\mathcal{A}$ denotes the set of active firms with $q_i^* > 0$. This demonstrates that as n grows, high-cost firms (larger β_i) are driven out, and the market stabilizes with only the most cost-efficient firms producing in equilibrium.

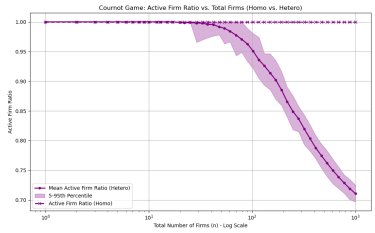
B. 1 to N players in Cournot Game with Homogeneous Cost

Assuming that all firms share an identical cost structure ($\beta = 0.5$), our simulation results, illustrated in Figure 3, reveal that the market outcomes evolve as the number of firms increases from a monopoly to perfect competitive market. Individual firm output declines while total market output increases and converges to the limit $\alpha_d - \beta_d\beta$. The equilibrium price falls steadily toward the marginal cost β, reflecting the dilution of market power as competition intensifies.

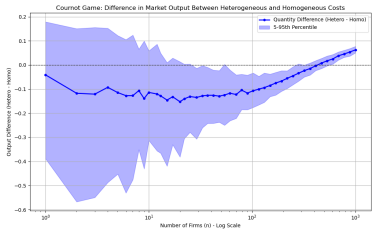
Figure 4 illustrates that the welfare outcomes change significantly with market entry. Consumer surplus expands as prices drop, whereas total producer surplus decreases, eventually approaching zero in a perfectly competitive market. Together, these results confirm that, in a homogeneous Cournot setting, increased competition leads to lower prices, diminished individual firm profits, and a reallocation of economic surplus from producers to consumers. Perfect competition achieves Pareto efficiency because resources are allocated such that no individual can be made better off without harming others. Total social welfare is maximized when price equals marginal cost.

C. Heterogeneous Cost and Firm Activity

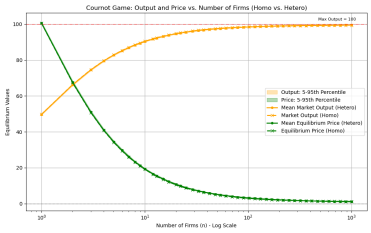
The introduction of heterogeneous marginal costs leads to asymmetries in output, pricing, and market share of each firm. Firms with lower marginal costs can sustain higher output and higher cost firms may exit the market if the equilibrium yields unprofitable quantities. The simulation assumes that firms marginal costs follow a lognormal distribution. This is motivated by the observation that production in many industries exhibit skewed distributions [8]. Majority of firms have the moderate cost level and a smaller fraction experience either lower or higher costs. The asymmetry also models the natural selection effect and barriers. This distribution also avoids negative values. The simulation selects randomly from the lognormal distribution ($\mu = \log(0.5), \sigma = 0.7$) to compute the equilibrium. Repeating this stochastic process 100 times results in a variety of outcomes. As competition intensifies, higher-cost firms are gradually forced out, reducing the proportion of active firms. Figure 5a illustrates this trend. Despite firm exits, total market output in heterogeneous cost settings remains close to that of homogeneous markets. Figure 5b demonstrates that lower-cost firms compensate for the production loss of higher-cost firms, stabilizing total output. This highlights how efficiency selection ensures that the most competitive firms dominate, preserving market efficiency even as competition intensifies. Figure 5c compares the market outcomes under homogeneous and heterogeneous cost structures. With a slight difference from the homogeneous case, the overall aggregated stability confirms the balanced market efficiency.



(a) Percentage of active firms

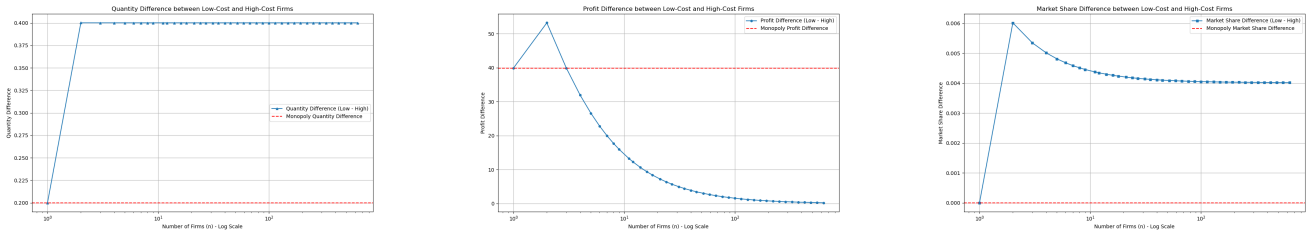


(b) Difference in total output



(c) Market outcome comparison

Fig. 5: Comparison of heterogeneous and homogeneous cost structures



(a) Output difference between low-cost and high-cost firms

(b) Price difference between low-cost and high-cost firms

(c) Market share difference between low-cost and high-cost firms

Fig. 6: Interaction of Participant Numbers and Cost Heterogeneity

D. The Interaction of Participant Numbers and Cost Heterogeneity

Differences in marginal costs create asymmetries among firms. In a duopoly, a lower-cost firm can produce at a higher quantity and thereby capture a larger market share in equilibrium. Intuitively, a cost-advantaged firm can undercut its rival by expanding output until its marginal profit reaches zero—a threshold that is achieved at a larger output level for the efficient, low-cost firm (as demonstrated in Section IV).

When we extend this analysis to an n-firm setting, the impact of cost heterogeneity becomes even more nuanced. Although lower-cost firms continue to produce more than their higher-cost competitors, the entry of additional firms increases the total market output and drives prices closer to marginal cost. As a result, while the absolute output of a low-cost firm rises due to a larger overall market, its relative market share diminishes as competition intensifies. Conversely, high-cost firms face a steeper competitive challenge, which forces them to cut back on production—and may even lead to their exit from the market.

The following simulation assumes that, only one low-cost firm ($\beta = 0.1$) and one high-cost firm ($\beta = 0.9$) exist in the market, the rest are medium-cost firms at ($\beta = 0.5$), except for the monopoly cases.

- Shown in Figure 6a, the equilibrium quantity difference between the high-cost and low-cost remains constant, indicating that the absolute production gap is determined by the cost and independent from the number of firms.
- Although a low-cost firm enjoys a significant profit advantage in a duopoly, as illustrated in Figure 6b the profit gap is narrowed alongside with the profit margin being compressed closer to the marginal cost when towards the perfect competition.
- Similarly, according to Figure 6c, while the low-cost captures a larger absolute output, its relative market share declines as n grows, the difference between the two also diminishes.

Therefore, the increasing number of agents also modulates the competitive advantage by cost heterogeneity. Though the low-cost firm retains a fixed production advantage, the benefits of doing so in terms of profit and relative market share are eroded by intensified competition.

VI. Collusion in Cournot Game

The first-order derivative of equation (8) w.r.t. q_2 is:

$$\frac{\partial \pi_1}{\partial q_2} = -\frac{q_1}{\beta_d} < 0 \tag{27}$$

,which shows that as firm B increases its output, firm A's profit reduces. In the other words, for smaller outputs of firm B, firm A's maximal profit is higher. The dash lines in Figure 7 below show both agents' iso-profit curve corresponding to the profit they make in an equilibrium, respectively.

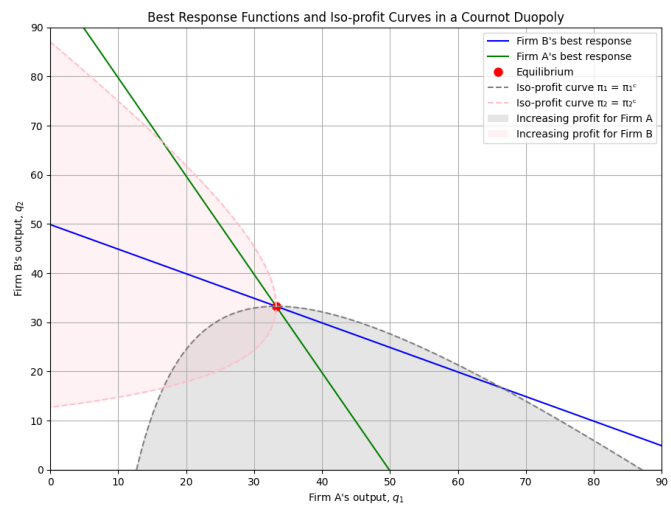


Fig. 7: Joint Profit Maximization in Cournot Duopoly

Any of the points in the lens-shaped intersection area is the set of outputs where both firms' profits are higher than in the Cournot Nash equilibrium.

In a collusive duopoly, firms maximize joint profits. Both firms collaborate as a single monopoly and split the monopoly output. The total profit for both firms under collusion is the total revenue (TR) minus their respective costs.

$$\Pi = (\frac{\alpha_d - Q}{\beta_d})Q - \beta_1 q1 - \beta_2 q_2 \tag{28}$$

, substituting equation (5)

$$\Pi = \frac{\alpha_d(q_1 + q_2)}{\beta_d} - \frac{(q_1 + q_2)^2}{\beta_d} - \beta_1 q1 - \beta_2 q_2$$

Taking the first derivatives w.r.t. q_1 and q_2 and setting them to zero,

$$\frac{\partial \Pi}{\partial q_1} = \frac{\alpha_d}{\beta_d} - \frac{2(q_1+q_2)}{\beta_d} - \beta_1 = 0$$

$$\frac{\partial \Pi}{\partial q_2} = \frac{\alpha_d}{\beta_d} - \frac{2(q_1+q_2)}{\beta_d} - \beta_2 = 0$$

Substract the second equation from the first yields

$$\beta_1 = \beta_2$$

Optimal (monopolistic) allocation of output among two firms requires equal marginal costs. Assuming that all firms share an identical cost structure ($\beta = 0.5$), the two firms will split the output equally. In a monopoly setting, substituting equation (4), the cost $C(Q) = \beta Q$, the joint profit (Π_M) can be expressed as:

$$\Pi_M = \frac{\alpha_d Q}{\beta_d} - \frac{Q^2}{\beta_d} - \beta Q \tag{29}$$

Taking the first-order derivative w.r.t. and setting it to zero yields the optimal monopoly quantity:

$$Q^M = \frac{\alpha_d - \beta\beta_d}{2} \tag{30}$$

The maximal total profits will be:

$$\Pi^M = \frac{(\alpha_d - \beta\beta_d)^2}{4\beta_d} \tag{31}$$

Each firm in the collusive duopoly produces:

$$q^M = \frac{Q^M}{2} = \frac{\alpha_d - \beta\beta_d}{4}$$

Each firm generates profit:

$$\pi^M = \frac{\Pi^M}{2} = \frac{(\alpha_d - \beta\beta_d)^2}{8\beta_d}$$

However, there will be an incentive for both firms to break the cartel and raise their profit by raising their quantity above the agreed level because collusive quantity is below their reaction curve. Suppose Firm A knows Firm B will produce the quantity as per the cartel agreement, Firm A's best response can be derived by substituting q^M to equation (11):

$$q_1^{M*} = \frac{3(\alpha_d - \beta\beta_d)}{8}$$

Firm A's profit will deviate to:

$$\pi_1^{M*} = \frac{9(\alpha_d - \beta\beta_d)^2}{64\beta_d}$$

,while firm B's profit will be reduced to:

$$\pi_2^{M*} = \frac{3(\alpha_d - \beta\beta_d)^2}{32\beta_d}$$

If both firms defect, the outcome will revert to Cournot equilibrium. A prisoner's dilemma structure can be used to analyse the problem of sustaining the collusion. The numerical results can be seen from Table V.

Defection is a strictly dominant strategy for both firms, therefore, the Nash equilibrium is for both firms to not abide by any collusive agreement.

		Firm B	
		Collude	Defect
Firm A	Collude	(2487.52, 2487.52)	(1865.64, 2798.46)
	Defect	(2798.46, 1865.64)	(2211.13, 2211.13)

TABLE V: Prisoner's dilemma in Cournot collusion

Hence, in a single-shot static game, the collusion are unlikely to be sustained under informal agreement. A contract can be used to ensure that each firm sticks to the agreed (collusive) output. Such contract is individually rational as the expected pay-offs (2487.52) from signing the contract are greater than their security values (the Cournot equilibrium level). However, another challenge to maintaining the collusive agreement is that a competitive fringe of firms will enter the market and drive the price down even when dominant firms have agreed on restricting production.

If firms have different cost structures, sustaining collusion becomes considerably more complex. Low-cost firm is less afraid of returning to Cournot equilibrium, as its equilibrium profits is relatively higher than their high-cost competitor, hence it has stronger incentives to increase profits by expanding output at lower marginal cost.

Differences in cost structures create difficulties in fairly distributing profits among colluding firms. The optimal production strategy for the collusion when forming monopoly is to let the low-cost firm to produce the majority of the output to minimize total costs. If their profit is proportionally to their output, a high-cost firm will receive a smaller share of the total profit, leading to an imbalance in profit distribution. To maintain collusion, additional mechanisms, such as side payments or quota agreements are necessary to ensure stability and fairness. The high-cost firm has incentive to deviate without such arrangements.

VII. Games in the real-world

Airlines' decision of increasing flight capacities on a specific route can be interpreted as a Cournot game when decisions are made simultaneously and independently. The quantity q that the airline can choose represents the transport capacity it offers. Although the ticket price is ultimately set by airlines based on the service offerings and classes, but we assume the market price are dynamic based on the total available seats. This can be modeled as a linear demand function using equation (1)

In a duopoly setting, from equation (27) , firm A's profit reduces when firm B sets massive capacities. If both firms aggressively set their flight capacities, the excess supply will result in a significant drop in market price, hurting profitability. Eventually, airlines had to adjust capacity, reducing some routes and frequencies, and returning prices to more profitable Cournot equilibrium levels. An example of such price war can be seen in Australia's domestic aviation market between Qantas and Virgin Australia between 2013 and 2014 [9].

In an oligopoly setting where more than two companies compete, airlines form alliances and coordinate capacities to reduce the number of independent Cournot competitors n, resulting in higher market prices as can be seen from Figure 3.

Regulators also closely examine alliances through a Cournot competition perspective to avoid excessive market power and higher fares for customers. Unlike typical Cournot collusion under informal agreement, which is hard to sustain, airline alliances have legally binding agreements. This often involves benefit sharing, which reduces competitive incentives among members.

References

[1] Jean Tirole. *The Theory of Industrial Organization*. MIT Press, 1988.

[2] Tejvan Pettinger. *Duopoly*. Economics Help, June 2020. URL: https://www.economicshelp.org/blog/glossary/duopoly/.

[3] Heinrich von Stackelberg. *Market Structure and Equilibrium*. Ed. by Mokhtar S. Bazaraa. English translation. Springer, 2011.

[4] John F. Nash and Sylvia Nasar. "Equilibrium Points in N-Person Games". In: *The Essential John Nash*. Ed. by Sylvia Nasar. Princeton University Press, 2016, pp. 49–50.

[5] Andrew F. Daughety. "Reconsidering Cournot: The Cournot Equilibrium Is Consistent". In: *The RAND Journal of Economics* 16.3 (1985), pp. 368–379. URL: https://www.jstor.org/stable/2555561.

[6] Drew Fudenberg and Jean Tirole. *Game Theory*. Cambridge, Massachusetts: MIT Press, 1991.

[7] Investopedia. *Producer Surplus*. Accessed: 2025-03-20. 2024. URL: https://www.investopedia.com/terms/p/producer_surplus.asp.

[8] John Sutton. *Market Structure: Theory and Evidence*. MIT Press, 2007.

[9] Guardian staff reporter. *Qantas says it won't add new capacity, ending price war with Virgin Australia*. the Guardian, May 2014. URL: https://www.theguardian.com/business/2014/may/21/qantas-says-it-wont-add-new-capacity-ending-price-war-with-virgin-australia (visited on 03/19/2025).

Declaration of the Use of Generative AI

Generative AI tools have been used to assist in the development of the code. Models include: Open AI GPT 4.5, o3-mini-high, and Claude Sonnet 3.7. It has been used to generate some initial code snippet for figures shown in this report. The code had been reviewed and modified to fit for various equations and visual components. It was also used when checking and debugging with the code written by us.

The report was written by us, Grammarly and GPT 4.5 were used for grammar checking, text enriching and rewording. It also supported in formatting math formula and LaTeX debugging.